

ORIGINAL ARTICLE

Enhanced deep-style interpreter for automatic synthesis of annotated medical images

Marcos Sergio Pacheco dos Santos Lima Junior¹ 

Juan Miguel Ortiz-de-Lazcano-Lobato^{1,2,3}  · José David Fernández-Rodríguez^{1,2,3} 

Ezequiel López-Rubio^{1,2,3} 

Received: 26 July 2024 / Accepted: 9 July 2025 / Published online: 4 August 2025

© The Author(s) 2025

Abstract

Creating an annotated medical image dataset is challenging and traditionally reliant on labor-intensive manual annotations. Additionally, these datasets often present substantial imbalances regarding sensing devices, class of medical disorders, and patient ethnicity and phenotype. Recently, there has been a research interest in mitigating these issues by employing data augmentation with generative models. However, the quality of images and semantics in medical image datasets are critical for computer vision tasks such as image segmentation. This paper presents DatasetGAN2-ADA, which aims to mitigate these difficulties by presenting an innovative deep-style interpreter robust against anomalous synthesis and designed to automate annotated image generation entirely. By leveraging the capabilities of StyleGAN2-ADA with an improved architecture of DatasetGAN and an enhanced execution framework integrated with an anomaly detector based on custom features, we propose a combined strategy for eliminating flawed synthetic images and masks. Furthermore, we propose exploiting image projections and preexisting semantics, eliminating the need for manual annotations to train our deep-style interpreter. The experimental results obtained with a magnetic resonance image (MRI) dataset demonstrate that DatasetGAN2-ADA is strongly effective in improving the efficiency and quality of synthetic generation, rejecting the synthesis of a substantial amount of low-quality images and masks. Then, an extension of this method is evaluated for detecting anomalous latent vectors *a priori* of the image synthesis, achieving up to 95.24% precision and illustrating its compelling potential for practical applications in medical imaging.

Keywords Deep learning · Generative adversarial network · Semi-supervised learning · Semantic segmentation

1 Introduction

Deep learning has shown numerous benefits in medical imaging applications in recent years, encompassing varied tasks such as disease diagnosis, lesion segmentation, and dataset augmentation [1, 2]. However, training a deep learning model for image analysis usually requires a substantial amount of images. If the training datasets do not contain enough images, models might overfit, which could significantly reduce their generalization capability [3]. This phenomenon is observed for image analysis in general, but it is especially important when dealing with medical tasks, which usually rely on scarce datasets and demand to be robust against data noise and distortions in order to achieve a reliable performance across multiple populations and different imaging sensors [4–6].

In addition, the images should have semantic information to train such deep learning models in a supervised or semi-supervised workflow. In the case of medical applications, it could be, for example, image labeling, key-point



information, bounding boxes, or pixel-wise segmentation. However, extracting such semantics is usually a very demanding process requiring a substantial amount of time from experts to achieve an acceptable level of quality.

Besides these challenges, medical image datasets are expected to be strongly imbalanced because images of healthy subjects usually appear much more frequently than those featuring medical disorders. Moreover, if the images of diseased patients are further subdivided into classes corresponding to the type of pathological condition observed, the dataset will become even more imbalanced because some medical conditions have a higher probability of occurrence than others. Still, other factors, such as the types of medical imaging sensor devices, patient ethnicity, and patient phenotype, make it difficult to acquire a balanced dataset with enough samples to represent the diversity required.

1.1 Related work

Therefore, several strategies have recently been explored to minimize these obstacles in medical imaging. Yu et al. [7], for example, reported an increasing tendency in publications to employ transfer-learning to solve data scarcity in medical images in this decade. Pointing toward the same direction, Ayana et al. [8] overview the benefits of multistage transfer-learning for solving medical image scarcity issues across several medical conditions, such as breast cancer, pulmonary diseases, neurological disorders, skin cancer, and others.

Other studies explored the employment of medical image augmentation to address this problem [9–11], where special concerns have been addressed about the current limitations of using augmentation techniques for brain magnetic resonance (MR) imaging [9].

In this context, using generative models to provide dataset augmentation has gained a lot of research interest as a promising technique [12]. Variational autoencoders (VAEs) represent one of the earliest generative propositions and have been applied to generate medical data with many applications [13], such as breast cancer detection [14]. On the other hand, generative adversarial networks (GANs) have been used to augment training datasets for thyroid tomographies [15], patch-based segmentation of retinal tomographies [16], brain MRIs [17], and integrated with VAEs for breast cancer detection [18], among others [10, 13, 19]. Finally, diffusion models represent one of the most novel generative paradigms and have been used for many applications [13], including image generation to improve segmentation [20] and classification [21] tasks. While diffusion models are currently much more computationally expensive than GANs and VAEs, they usually generalize better, leading to higher quality in generated images. However, their latent space still needs to be explored, which makes them a more challenging choice for image projection and editing when compared to GANs [22].

Regarding MR imaging specifically, the employment of augmentation strategies based on generative networks still remains an open challenge due to issues such as domain adaptation, generalization, computational complexity, and reliability [9, 23]. These issues are aggravated when dealing with semantic segmentation of brain strokes due to factors such as lesion heterogeneity, indistinct margins, co-occurring pathology, artifacts, and intensity inhomogeneity [24].

In this work, we present DatasetGAN2-ADA, which addresses these challenges by assessing the augmentation of a dataset that contains magnetic resonance images and segmentation masks of human brains. DatasetGAN2-ADA is based on DatasetGAN [25], with a more robust framework and an improved backbone that employs StyleGAN2-ADA [26]. Additionally, it includes an anomaly detector based on an innovative custom feature extractor combined with a semantics filter for a more robust synthesis, the novel embedding of image projections to train DatasetGAN2-ADA without the need for image annotations, an *a priori* latent vector classifier that prevents the generation of anomalous images in the first place, and an enhanced execution framework.

The main propositions brought by this work are presented in Figure 1 and are explained in detail in the following sections. Where Sect. 2 describes the research methodology and experiment framework, Sect. 3 expatiates about the experiments conducted, Sect. 4 brings the results and discussions, and Sect. 5 leads to the conclusions.

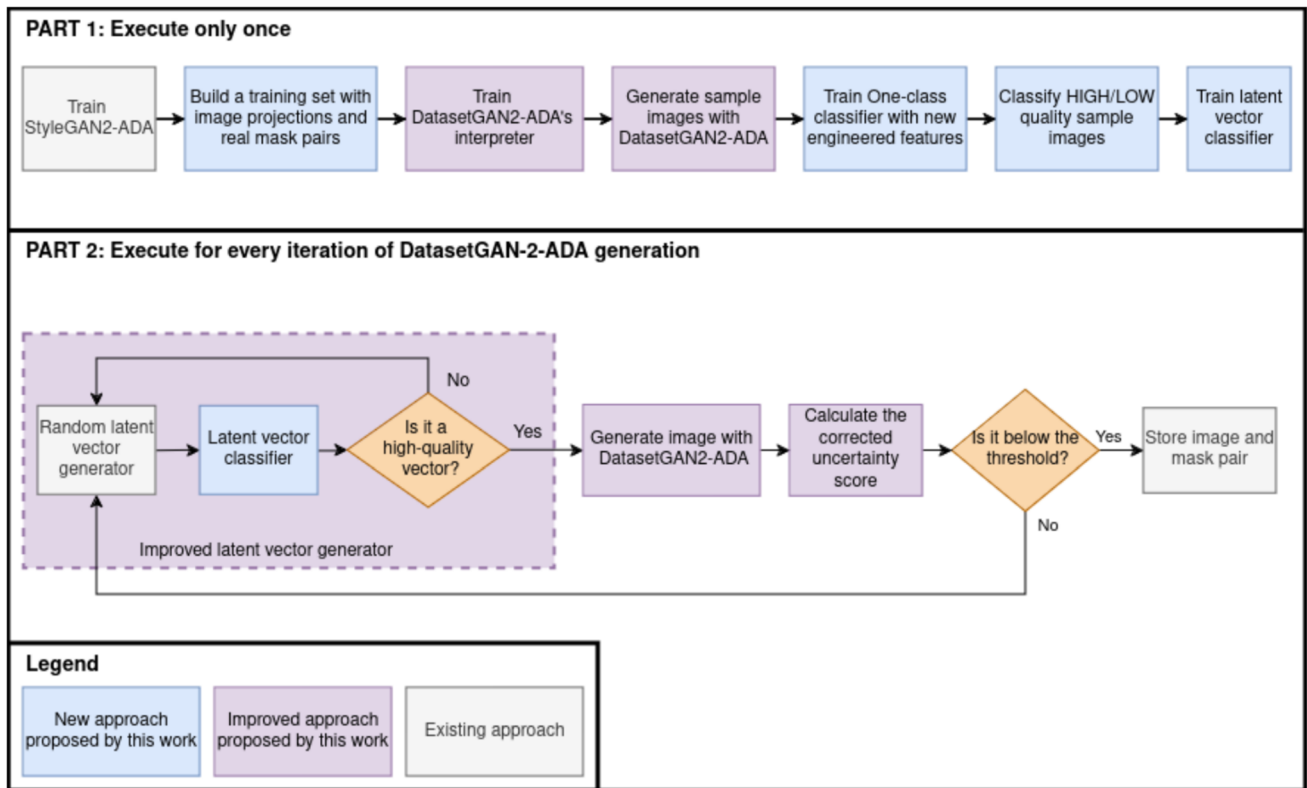


Fig. 1 Novel workflow proposed by DatasetGAN2-ADA

2 Methodology and experiment framework

This work uses the ATLAS 2.0 dataset [27], containing MRIs of the human brain of 655 individuals. The dataset provides images previously registered and normalized, combined with segmentation masks containing two classes: lesion and non-lesion. Fourteen scanner models from different manufacturers were used to generate the MRI files, each storing a 3D volume scan of an individual's head. The MRI files are voxelized based on a system of orthogonal axes (left-right, anterior-posterior, and inferior–superior), which might also be interpreted as a sequence of 2D slices along one of the orientation axes.

As the MRI brain scans in this dataset are already registered, the scans are expected to contain similar anatomical locations of the brain at the same slice index in any axes when compared across different subjects. Thus, it was decided to separate the MRIs into groups of 20 neighboring slices in the inferior–superior axis: 0–19, 20–39, 40–59, ..., 140–159, and 160–179. The slices 180 to 189 were discarded because they mostly feature void space around the top of the skull, devoid of brain tissue. Such slice grouping was beneficial because it provided sets with enough images for the experiments, as it is further explained in Sect. 2.1.1.

After the grouping, all image slices were processed to have a resulting shape of 256×256 pixels. For that, the 197×233 original inferior–superior slices were centered in a 256×256 frame, and the background of the frame was filled with the minimum pixel intensity found in the outer region of the brain of the original images.

Finally, the group containing the slice indexes from 80 to 99 in the inferior–superior axis was chosen to conduct the experiments described in the following sections because it contains a central portion of the brain, usually including the region where the circumference of the head is maximal. Hence, the final dataset employed in the experiments contains 20 MRI slices of 655 individuals, summing up 13,100 image and mask slice pairs, each with 256×256 pixels.

2.1.1 StyleGAN2-ADA backbone

Considering the use of a relatively smaller dataset, we propose to make DatasetGAN2-ADA compatible with StyleGAN2-ADA, which uses an adaptive discriminator augmentation (ADA) pipeline that tunes the augmentation strength dynamically based on the degree of overfitting, enabling the training of the discriminator without leaking the augmentation pattern. Such modification significantly improves the performance of style-based generators on smaller datasets. While StyleGAN2 has good performance with datasets containing 10^5 to 10^6 images, StyleGAN2-ADA allows good performance in datasets with 10^4 images, which is the order of magnitude of the dataset used in this study [26].

2.1.2 Improved Style Interpreter architecture

Originally, the style interpreter was an MLP with 3 fully connected layers having a ReLU activation followed by a batch normalization between each of the fully connected layers. We propose to make this architecture more robust by adding two modifications. The first is the replacement of the ReLU activations by SiLU activations, which is known to provide a better self-stabilizing property by acting as an implicit regularizer to avoid large weights [32]. The second is the addition of a Lasso regularization to the training scheme, aiming to prevent overfitting in a scenario with limited training images and where a subset of the 5888 style features may not be useful for the segmentation of brain lesions.

2.1.2.1 Style interpreter assessment framework A cross-validation experimentation framework was designed to evaluate the performance of the style interpreter with and without the changes proposed. In this scheme, the dataset was split into 8 folds and trained for 15 epochs with the same feature maps used in the experiments described in Sect. 3. At the end of each epoch, the mIoU (mean intersection over union) was calculated and stored. Then, at the end of the cross-validation, the mean mIoU over the folds was computed per epoch. Table 1 displays the maximum mIoU achieved for each network configuration along the 15 epochs. It is possible to observe that the configuration containing the SiLU activations combined with L1 regularization obtained the best mIoU scores, proving the advantages of the new architecture proposed for the DatasetGAN2-ADA's style interpreter.

2.1.3 Leveraging latent vector projections to avoid manual annotations

Recently, multiple initiatives have studied different approaches dealing with the manipulation of StyleGAN latent vectors for the generation of high-fidelity image projections [30, 33, 34]. The main idea is to find latent vectors in StyleGAN's mapping network, which correspond to target images, enabling a more controlled synthesis of synthetic images.

We benefited from these studies to propose a high-fidelity methodology for generating image projections. Then, we employed these projections in a proposition that eliminates the manual annotations for training DatasetGAN's interpreter.

Table 1 Ablation studies of the proposed modifications to the style interpreter architecture, where the baseline architecture is the original implementation introduced by DatasetGAN

Setup configuration	Maximum mIoU achieved
Baseline architecture	0.9817
Replacing ReLU by SiLU	0.9820
Baseline + L1 regularization ($\lambda = 10^{-8}$)	0.9823
Replacing ReLU by SiLU + L1 regularization ($\lambda = 10^{-8}$)	0.9825

2.1.3.1 Image projection framework The mapping network of the StyleGANs originally uses input latent vectors in Z space to generate output latent vectors in W space. Then, the latent vectors in W are concatenated twice to feed each style block in StyleGAN's synthesis network [28]. After this concatenation, the resulting higher-dimensional form of the latent vectors belongs to a latent space named W^+ , which has been demonstrated to be the most effective space for high-fidelity image projections using StyleGANs [33]. This process is illustrated in Figure 3.

The key idea behind an image projection using W^+ space is to find a vector $w^+ \in W^+$ such that when w^+ is used as an input to a pretrained StyleGAN's synthesis network, it generates as an output an image that aims to be identical to a targeted image.

In this work, we use a mixture of ideas from different publications [26, 33] to generate image projections with the required quality for our experiments. In summary, a VGG16 feature extractor ($f_{extractor}$) was used to compare the target images with the result of StyleGAN2-ADA synthesis through an iterative process in order to find the w^+ latent vector that best represents the target image. This process, which is computed through N iterations, is described in detail in Algorithm 1. It aims to find a value of w^+ that minimizes the distance $D(T, \tau)$ between the target image T and the synthetic image τ generated at each of its iteration by StyleGAN2-ADA. The search for the best w^+ was conducted by employing a gradient optimization with additive noise (η) because the use of noise has known benefits to improve the quality of the projections [34]. The predefined constants $C_1 = 0.05$ and $C_2 = 0.75$ are used to control the noise intensity in each iteration.

Algorithm 1 Find $w^+ = \arg \min_{w^+ \in W^+} D(T, \tau)$

```

1:  $w_0^+ \leftarrow w_{mean}^+ : w_{mean}^+ = \frac{1}{|W^+|} \sum_{w^+ \in W^+} w^+$ 
2:  $optimizer \leftarrow w_0^+$ 
3: for  $i = 1, 2, \dots, N$  do
4:    $\eta(i) \leftarrow [C_1 \sigma_{w^+} \max(0; 1 - \frac{i}{C_2 N})] \times \eta_{random}$ 
5:    $w_i^+ \leftarrow w_{i-1}^+ + \eta(i)$ 
6:    $\tau_i \leftarrow StyleGAN2ADA(w_i^+)$ 
7:    $D_i \leftarrow D_{euclidean}(f_{extractor}(T); f_{extractor}(\tau_i))$ 
8:    $loss_i \leftarrow D_i + \eta_{regularization}$ 
9:    $backpropagate(loss_i)$ 
10:   $w_i^+ \leftarrow step(optimizer)$ 
11: end for

```

2.1.3.2 Eliminating the dependency on manual annotations in DatasetGAN Manually annotating images is cumbersome, takes time, and might require the knowledge of a specialized professional in the subject. Although DatasetGAN can automatically generate images with its corresponding annotation pair, it must have several manually annotated images to train its interpreter properly. In this context, the manual annotation process is required because its interpreter requires training sets of corresponding latent vectors, images, and masks, which would only be feasible by synthesizing new images with DatasetGAN.

As this work benefits from using image datasets with annotations, we project the real images onto the latent space to find their corresponding latent vector following the process described in this Section. Then, we propose utilizing the projected images and their latent information together with the corresponding previously provided real masks to train DatasetGAN's interpreter. This way, manually annotating images to train DatasetGAN2-ADA's style interpreter is no longer required. This process is illustrated in Fig. 4.

Fig. 3 Representation of the Z , W and W^+ spaces in StyleGAN

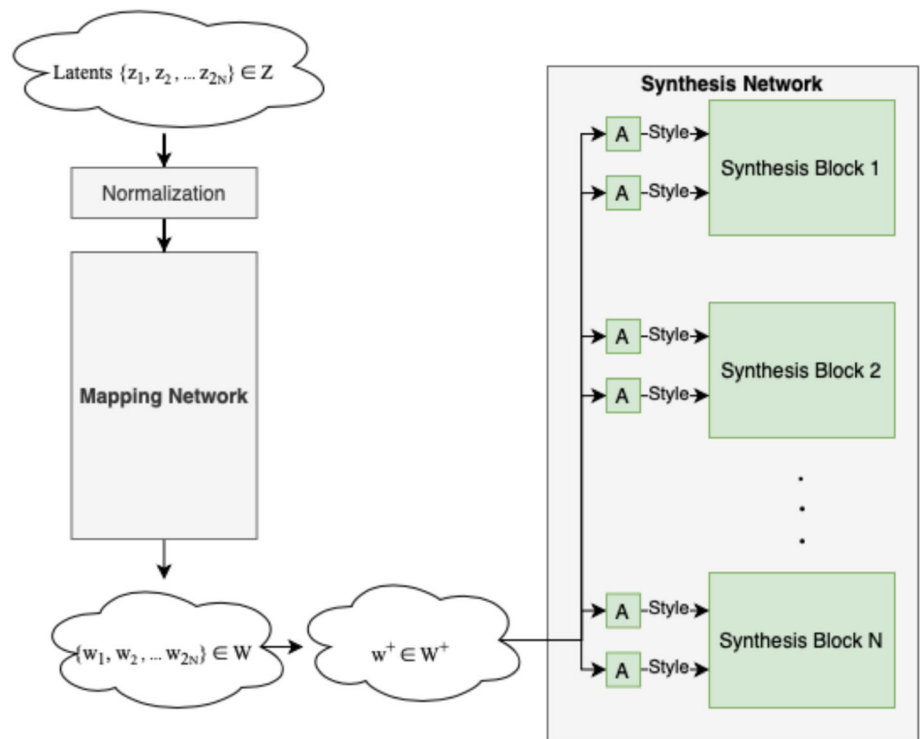
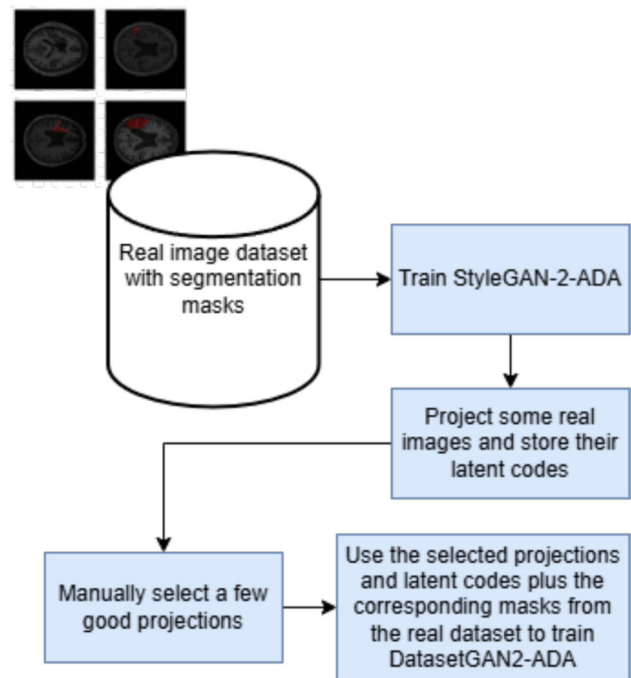


Fig. 4 Automation of the manual annotation process to train DatasetGAN2-ADA's interpreter using projections of real images previously annotated



2.1.4 Enhanced image synthesis

Another improvement proposed is in the image synthesis workflow, where a more robust latent vector generator is combined with an enhanced uncertainty score filtration. While DatasetGAN originally synthesizes images from any latent vectors generated by random choice, we propose to embed a pretrained latent vector classifier in the

random latent vector generator to predict whether the latent vectors to be used in image synthesis will yield high-quality images or not. Then, only the latent vectors that are predicted to yield high-quality images are used for image synthesis. This approach provides more control over image generation by preventing synthesizing low-quality images in the first place. Furthermore, it also optimizes the computational resources during image synthesis because predicting the quality of a latent vector is significantly less computationally expensive than synthesizing an image. The training of the MLP classifier is explained in detail in Sect. 4.3.

Additionally, we propose to improve the uncertainty score calculation of DatasetGAN during image synthesis by correcting an observed bias. This approach is detailed in Sect. 3.2, where we propose a heuristic method to correct the bias.

2.1.5 Revamped computational strategy

In addition, this work also used an improved computation strategy for the style interpreter training. In summary, the original method creates the feature maps used for training the style interpreter and stores all of them in the random access memory (RAM), requiring dozens (or even hundreds) of GB to run the code, depending on the amount and size of the annotated images. Our strategy overcomes it by using a different approach, which creates and stores the feature maps on the hard disk, allowing to load them into the RAM in chunks, one image at a time.

Furthermore, in the original DatasetGAN strategy, the feature maps are recomputed every time the interpreter is trained. Storing the feature maps onto the hard disk was also convenient because it avoided spending time and resources generating the same feature maps every time DatasetGAN2-ADA needs to be trained with the same dataset and different hyperparameters.

Table 2 compares the computational resources spent to train the style interpreter of DatasetGAN and DatasetGAN2-ADA using the same images to generate feature maps.

3 Experiments

After the improvements proposed for DatasetGAN2-ADA in Sects. 2.1.1, 2.1.2 and 2.1.5, the dataset described in Sect. 2 was used to train StyleGAN2-ADA. Then, its checkpoint was stored for the training of DatasetGAN2-ADA. After that, 24 image and mask pairs obtained by the process described in Sect. 2.1.3 were used to train the style interpreter of DatasetGAN2-ADA. Finally, we assessed the quality of 10,000 images generated by DatasetGAN2-ADA to conduct the experiments described in this section.

The first goal of the experiments was to understand whether or not these improvements would enable the synthesis of high-quality images and mask pairs for the particular case of MRI of human brains.

Table 2 Comparison of resources spent to train the style interpreter of DatasetGAN and DatasetGAN2-ADA from scratch using 24 annotated images with 256×256 pixels. The experiments were executed in the same computer with an Intel(R) Core(TM) i9-10900X CPU @ 3.70GHz with 20 cores, using 2 NVIDIA GeForce RTX 4090 GPUs

	DatasetGAN	DatasetGAN2-ADA
RAM used	47.8 GB	6.8 GB
GPU memory used	4.8 GB	2.6 GB
Execution time	1h20	7h20

3.1 Investigation of the root cause of the low-quality image synthesis

After qualitatively analyzing the synthesized MRI slices, it was possible to observe that several image and mask pairs show low quality. Figure 5 brings a small subset of the generated images to exemplify it. This observation is compatible with other studies [35] that investigated the use of DatasetGAN for medical images and expatiated about a portion of low-quality synthesis. For these reasons, this work also presents a strategy to automatically filter out the low-quality images targeting to end up with a superior-quality synthetic dataset.

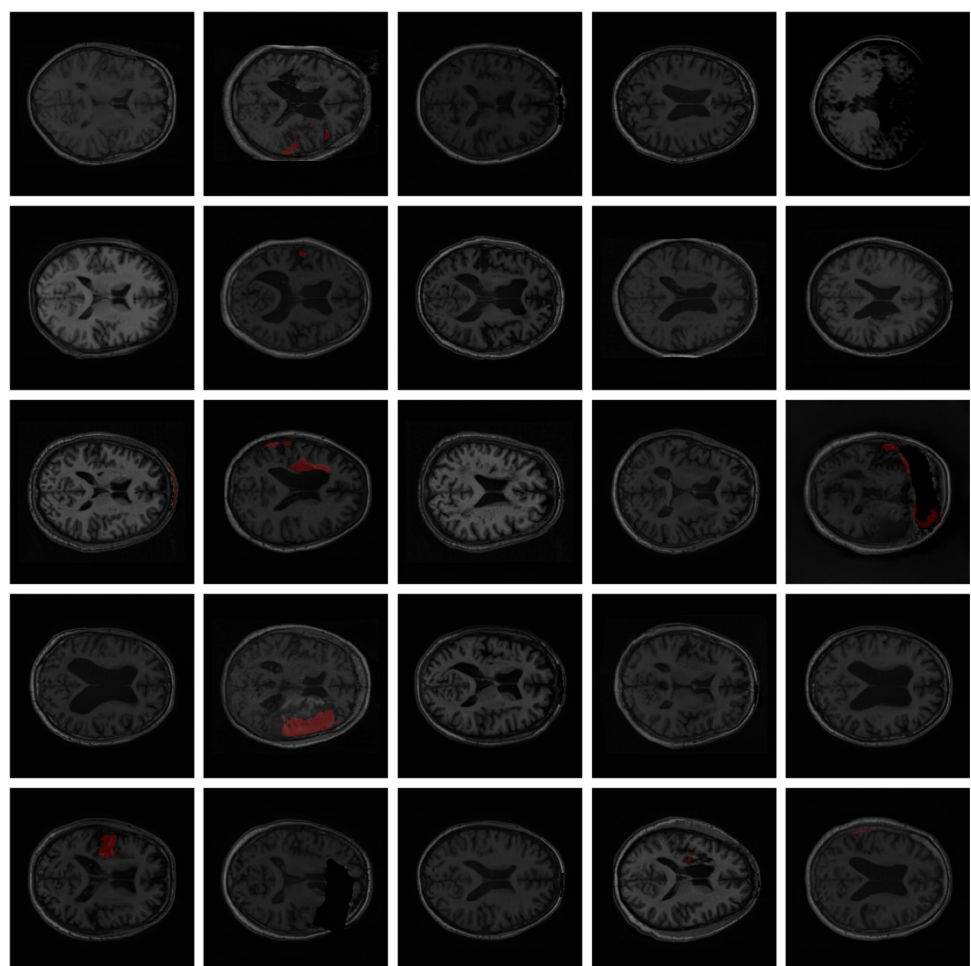
3.1.1 Separating the causes of the low-quality synthesis

It was observed that some of the images that presented good quality had a poor segmentation mask, while others showed low quality independently of its mask. Thus, we hypothesize that more than one root cause is behind the cases with inferior synthesis quality. Given that, we propose splitting the image filtering strategy into two approaches: focusing on the quality of the masks synthesized by DatasetGAN (A) and the quality of the images generated by StyleGAN2-ADA (B).

A - Focus on image segmentation: by correcting an observed bias in DatasetGAN's uncertainty score (*UncScore*) metric and filtering out the images with high *UncScore* values.

B - Focus on image synthesis: by filtering out the anomalous images using extracted features and performing anomaly detection.

Fig. 5 Random sample of synthetic images generated by DatasetGAN after the initial improvements proposed in Sect. 2



Approach A should be easier to tackle as the uncertainty score calculated after the pixel classification is performed by the ensemble of classifiers in the style interpreter and is already provided by the original DatasetGAN implementation [25]. However, we noticed that the imbalance of the classes in the image segmentation task produces a bias in the uncertainty score value. Thus, we propose a method for correcting this bias and setting a threshold for filtering out images with poor segmentation in Sect. 3.2.

Conversely, Approach B is not straightforward because no procedure for automatic classification of the low-quality images was previously determined, nor were metrics that could be used to classify the images into high or low quality. We explore the details of these methods proposed for removing anomalous images in Sect. 3.3.

3.2 Approach A: handling poor segmentation masks

The original DatasetGAN implementation employs the Jensen–Shannon (JS) divergence to calculate the uncertainty of using the ensemble of classifiers to attribute a class to each pixel during the image synthesis process. Then, it chooses the 10% of the pixels for which the JS divergence is higher and averages their JS divergence scores. The result is what DatasetGAN’s work entitles as the uncertainty score of each image.

This approach seems to work well on a multi-class segmentation task where the number of pixels corresponding to each class is somewhat balanced in general. However, our task is to segment brain lesions in MRI slices, which is strongly imbalanced because not all layers contain lesions and most of the layers with lesions has the vast majority of the pixels belonging to the no-lesion class.

We observed that the uncertainty of the pixels tends to be higher in regions around the lesions. As sometimes the region of the lesion corresponds to much less than 10% of image pixels, images with bigger lesions tend to have a higher uncertainty score, as shown in Figure 6. This way, choosing a constant uncertainty score threshold to filter out poor image segmentation tends to be complacent with images containing small portions of the lesion class segmented with low quality and filter out images with big lesions for which the segmentation is acceptable.

To solve this problem, we empirically derived a heuristic formulation for computing the bias of the uncertainty score, as shown in Equation 1.

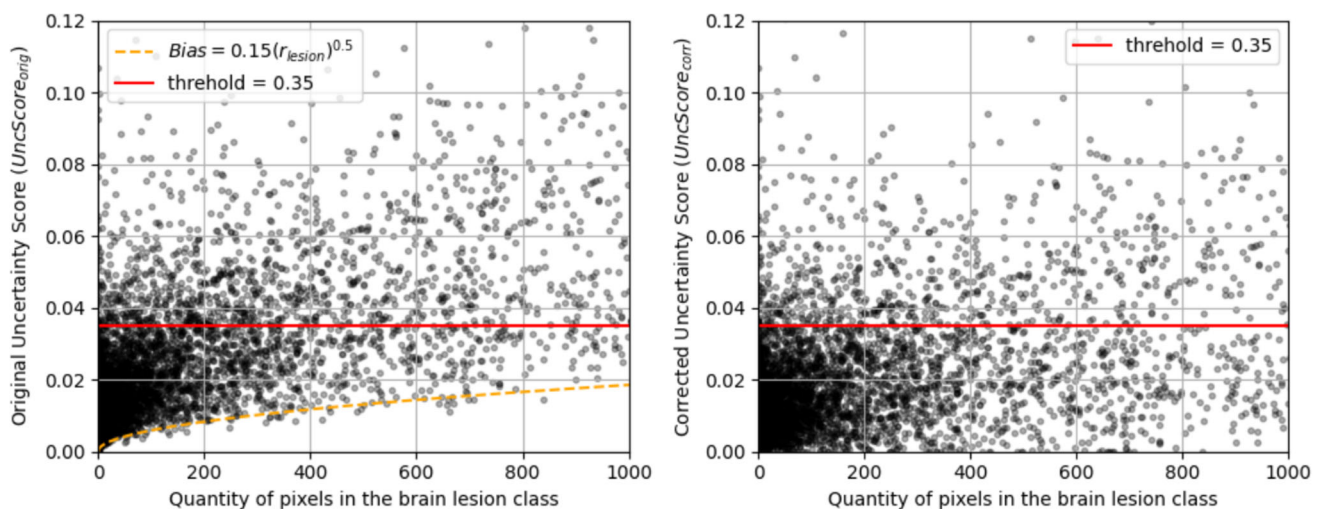


Fig. 6 Uncertainty score correction. On the left the scatter plot of the uncertainty scores of the synthesized images alongside the perceived bias and an uncertainty score threshold of 0.035. On the right the a scatter plot of the corrected uncertainty scores and the same threshold of 0.035

$$bias = 0.15(r_{lesion})^{0.5} \quad (1)$$

Where r_{lesion} corresponds to the ratio of the image pixels in the brain lesion class. Then, in Equation 2, we used this bias for correcting the uncertainty score so that we might use a constant threshold without being prejudiced by the size of the brain lesion.

$$UncScore_{corr} = \arg \max\{0, \quad UncScore_{orig} - bias\} \quad (2)$$

Where $UncScore_{corr}$ is the corrected uncertainty score and $UncScore_{orig}$ refers to the original DatasetGAN's uncertainty score without any corrections.

3.3 Approach B: handling anomalous image synthesis

The first step in finding a solution for approach B was to extract features from the images that could provide relevant information about their perceptible quality. Extracting those features is an initial step before training a machine learning model to classify whether the images have low or high quality.

3.3.1 Engineering new features

For this purpose, eight custom features were conceived to detect a few characteristics that were observed to be more frequent in low-quality images after a thorough observation. Those characteristics include erratic brain shapes and sizes, failed image syntheses, unconventional positions, and orientations, artifacts outside the heads, images with missing information, blur, and unusual or varying brightness. A few examples are shown in Figure 7.

Feature I: brain area

This feature aims to detect unusual small or big heads. The area of the head was determined by the following procedure:

- i - Calculate the mean MRI of the real dataset.
- ii - Determine the region corresponding to the mean MRI head by calculating the convex hull of the mean MRI slice with a predetermined threshold for the pixels' intensity as illustrated in Figure 8.
- iii - Considering that the brain images were previously registered, normalize the image only in the region corresponding to the head in the mean MRI. This step is done because sometimes the synthetic images have a noisy background that imitates some images in the real dataset. In those cases, the intensity of the noisy pixels might be relatively high compared to other images. Thus, if it is not done, the individual convex hull computation in the following step has a higher chance of wrongly including an area corresponding to the image background due to these relatively high-intensity noisy pixels.
- iv - Detect the pixels above a predetermined threshold in the image that was partially normalized.
- v - Compute the convex hull of the image.
- vi - Calculate the area corresponding to the region inside the convex hull.

Feature II: Circumscribed rectangle orientation

This feature aims to detect the rotation angle of the head in the image and images with anomalous shape. It is calculated as follows:

- i - Compute the head contour using the convex hull computed in item v of the feature I.
- ii - Determine the minimum area rectangle that circumscribes the contour.
- iii - Find the orientation of the rectangle.

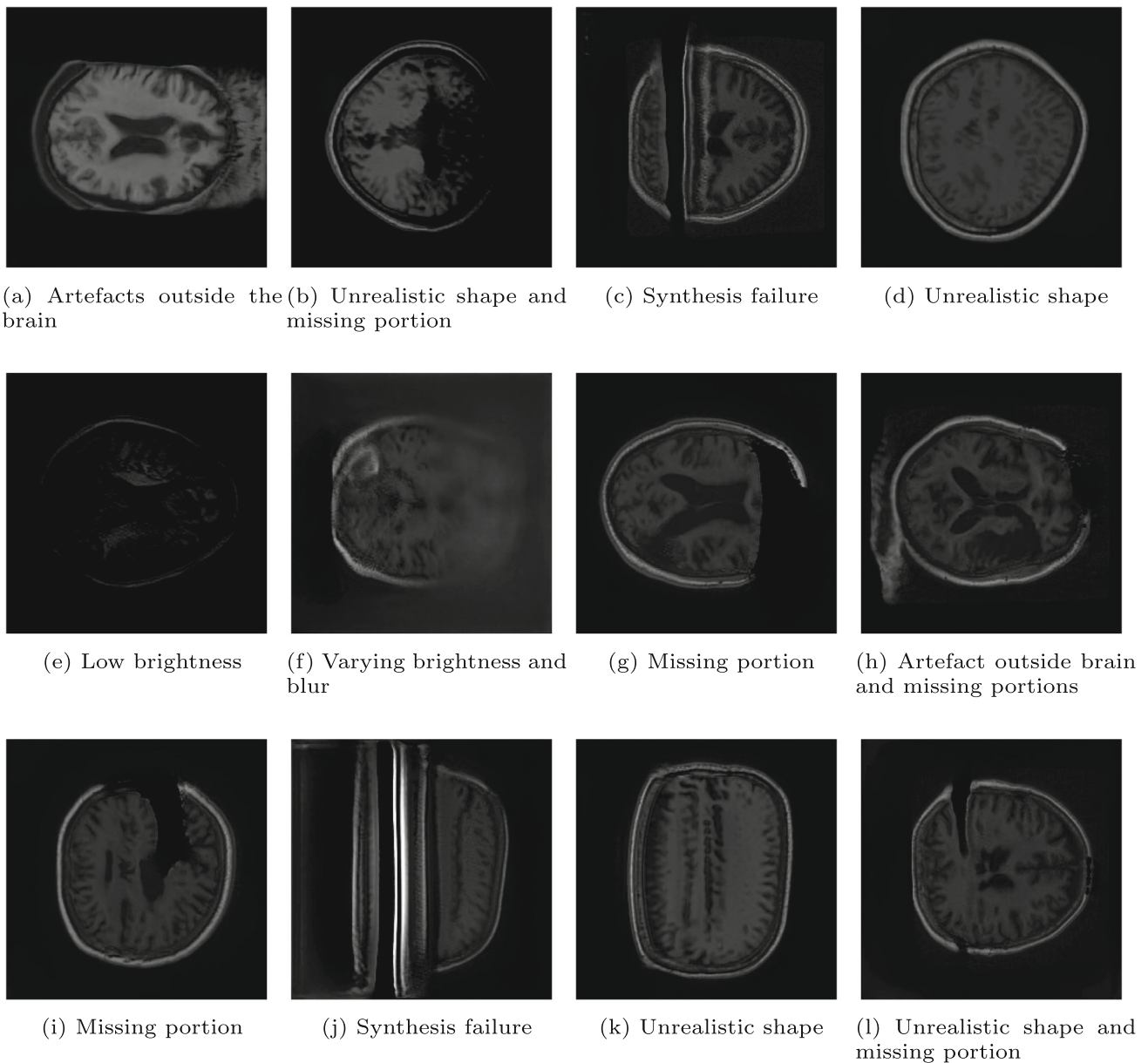


Fig. 7 Examples of problems in the image synthesis of StyleGAN2-ADA, including artifacts outside the brain, unrealistic brain shape, missing image portions, strong blur, brightness variation, low brightness, and synthesis failure

Feature III: Perimeter

This feature measures the perimeter of the head. Together with feature *I*, it helps to detect unusually large and small heads. Combining this feature with feature *I* allows the differentiation between normal-size heads with unusual rugosity in the head contour and brains with an unusual size. It is computed by measuring the contour length calculated in the item *i* of the feature *II*.

Features IV and V: Centroid X and Y

These two features aim to catch images where the head is not in the center or has unusually imbalanced centroids corresponding to both image slice axes. As the real dataset contains registered MRI brains, the positions of the heads' centroids are expected to be close to each other. These features are determined by calculating the centroid of the image's convex hull, calculated in step *v* of feature *I* (Fig. 9).

Feature VI: Intersecting ratio of the convex hull

Fig. 8 Mean image slice and its convex hull considering a threshold of 10 where the pixel intensity ranges between 0 and 255

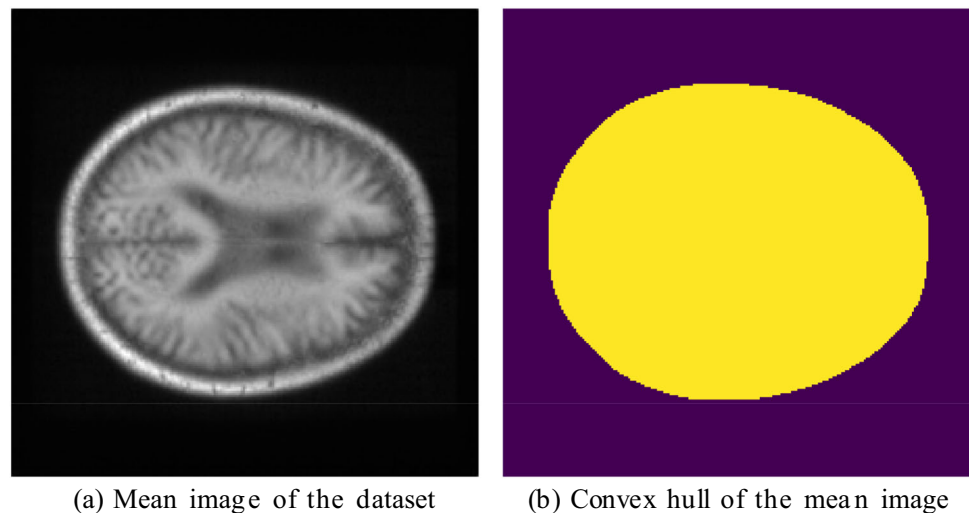
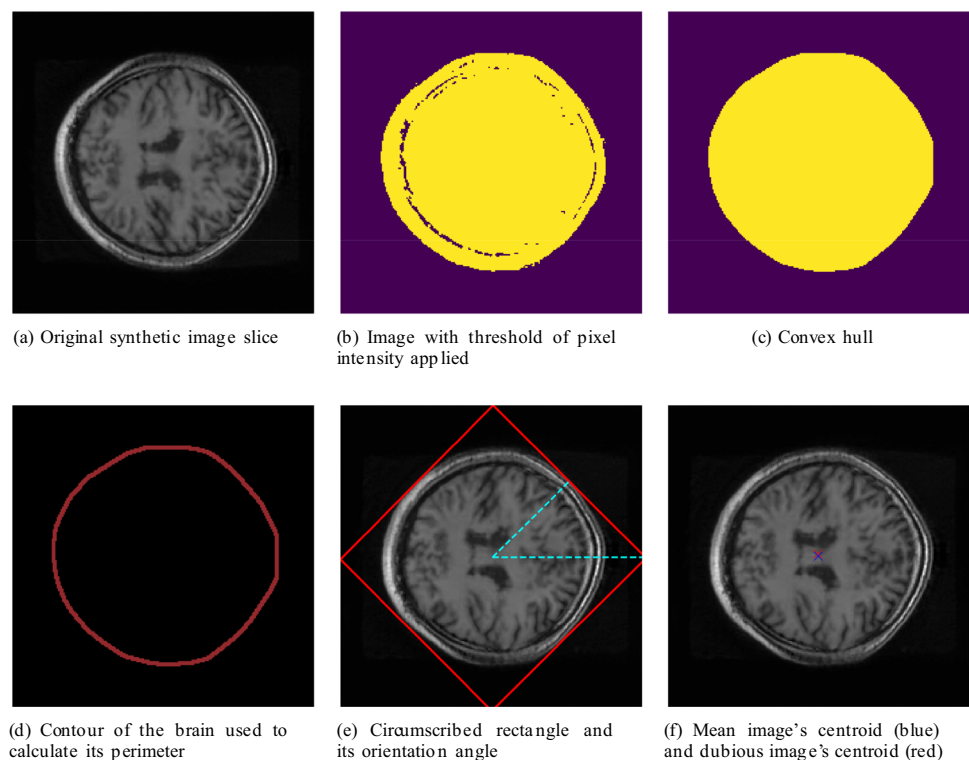


Fig. 9 Analysis of a dubious image slice: (a) original synthetic image slice; (b) image with a threshold of pixel intensity applied; (c) its convex hull; (d) contour of the brain used to calculate its perimeter; (e) circumscribed rectangle and its orientation angle; (f) mean image's centroid (blue) and dubious image's centroid (red)



This feature aims to check the ratio of the intersection of the synthetic images' convex hull with the convex hull of the mean real MRI. The goal is to evaluate if there are void cavities inside the head, but it also could indicate that the shape or position of the head is unusual.

Feature VII: Intersecting ratio of the background convex hull

This feature checks the intersection ratio of the area outside the convex hull of the synthetic images with the area outside the convex hull of the mean real MRI. This feature aims to catch artifacts outside the head, but unusual values may also indicate that the shape of the head is unusual.

Feature VIII: Circularity

The circularity [36] is a measure of similarity to a circle based on the variation of the radii of the head, where a value of 0 corresponds to a perfect circle. It is computed by finding the ratio between the standard deviation and the mean of the radii following the procedure described below:

- i* - Calculate the centroid of the head using the convex hull computed in item *v* of the feature *I*.
- ii* - Measure the distance between the centroid and the border of the head (radius) in all directions.
- iii* - Calculate the mean (μ) and standard deviation (σ) of the radii.
- iv* - Calculate the Circularity (*C*) as $C = \sigma/\mu$.

3.3.2 Anomaly detection using the engineered features

The next step was to normalize the engineered features and use them for training a model for automatically detecting which images correspond to high- and low-quality syntheses. As there are no labels for assessing the quality of the images and as the real image dataset contains only good examples, the approach chosen to solve this problem was to use a one-class classifier to detect the anomalies in the synthetic images. The classifier chosen was the Isolation Forest [37] because it does not require any threshold to determine the number of images chosen as anomalies and because it usually presents relatively low run time and high accuracy [38].

4 Results and discussions

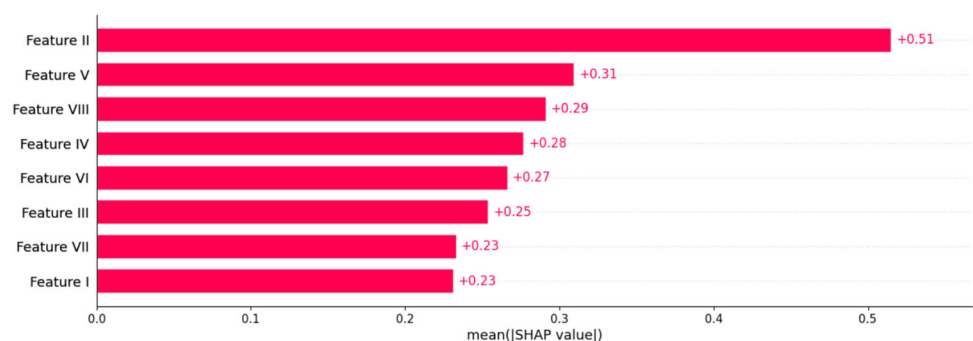
Firstly, the Isolation Forest one-class classifier was trained with all 13,100 real MRI slices using the features extracted with the custom feature extractor. Then, the same model was used to evaluate all the 10,000 synthetic MRI slices. As a result, 10.61% of the images were classified as anomalies, which we call the low-quality class. Finally, we evaluated how the features impact the one-class classifier to justify their usefulness.

4.1 Impact of the features in the anomaly detection

The influence of the features described in Sect. 3.3 was evaluated using quantitative and qualitative approaches. The first step was to employ SHAP (SHapley Additive exPlanations) [39] to estimate the importance of each feature in the overall result. Figure 10 shows a plot of the mean absolute SHAP value, indicating that all the features have a meaningful impact on the overall decision. Also, it is possible to observe that feature II (circumscribed rectangle orientation) has a significantly higher impact on the model's outcome.

To support this analysis, we evaluated the training of the one-class classifier with the same training data and hyperparameters but removing one of the features. This process was repeated for each of the features, and the results of the anomaly detection on the synthetic images were stored. Then, we counted the images classified as anomalies in the original training with all the features but not classified for each of the subsequent training runs.

Fig. 10 Mean absolute SHAP metric for each of the features



The results are shown in Table 3, where it is possible to observe that eliminating any of the features impacts the overall results, meaning that all features bring value to the one-class classifier. Also, similarly to the SHAP analysis, it is possible to notice that feature II has a significantly higher impact on the results.

Then, in Figure 11, a subset of the images listed in Table 3 is displayed to support the results qualitatively. It is possible to observe that among those images there are many samples containing the problems listed in Sect. 3.3.

4.2 Combining anomaly detection and uncertainty score threshold filtering

With these results, it became possible to evaluate the hypothesis formulated in Sect. 3.1.1 about the root cause of the image synthesis flaws. For that, we split the synthetic dataset into high- and low-quality classes according to the output of the Isolation Forest model and plotted the histograms of the corrected uncertainty score shown in Figure 12. The objective was to check whether combining a $UncScore_{corr}$ threshold filter with the high-quality class would benefit our method or not.

It is possible to observe that the images detected as anomalies tend to have higher uncertainty scores. This observation was verified by using the one-tailed Mann–Whitney U Test [40], where we assess if there is a statistically significant difference between the $UncScore_{corr}$ distribution in the high- and low-quality classes. This test's null hypothesis is that the $UncScore_{corr}$ distributions of the high-quality and low-quality classes are not different. Conversely, the alternative hypothesis states that the median of the $UncScore_{corr}$ of the low-quality class is higher than the median of the $UncScore_{corr}$ of the high-quality class, implying that the $UncScore_{corr}$ distribution is different.

As a result of the one-tailed Mann–Whitney U Test, a p-value of 3.62×10^{-43} was obtained, meaning that the null hypothesis is rejected with very high confidence, implying the distribution of the $UncScore_{corr}$ of the low-quality class is significantly more right-skewed.

On the other hand, both anomalous and good images classified by the Isolation Forest method have most of the images with $UncScore_{corr} < 0.02$, proving that only a low $UncScore_{corr}$ does not mean the image is not an anomaly. Reversely, there are low $UncScore_{corr}$ in both anomalous and good images classified by the Isolation Forest method, meaning that simply employing the one-class classifier alone is also not the best choice. Therefore, combining both strategies is expected to yield better results than using only one.

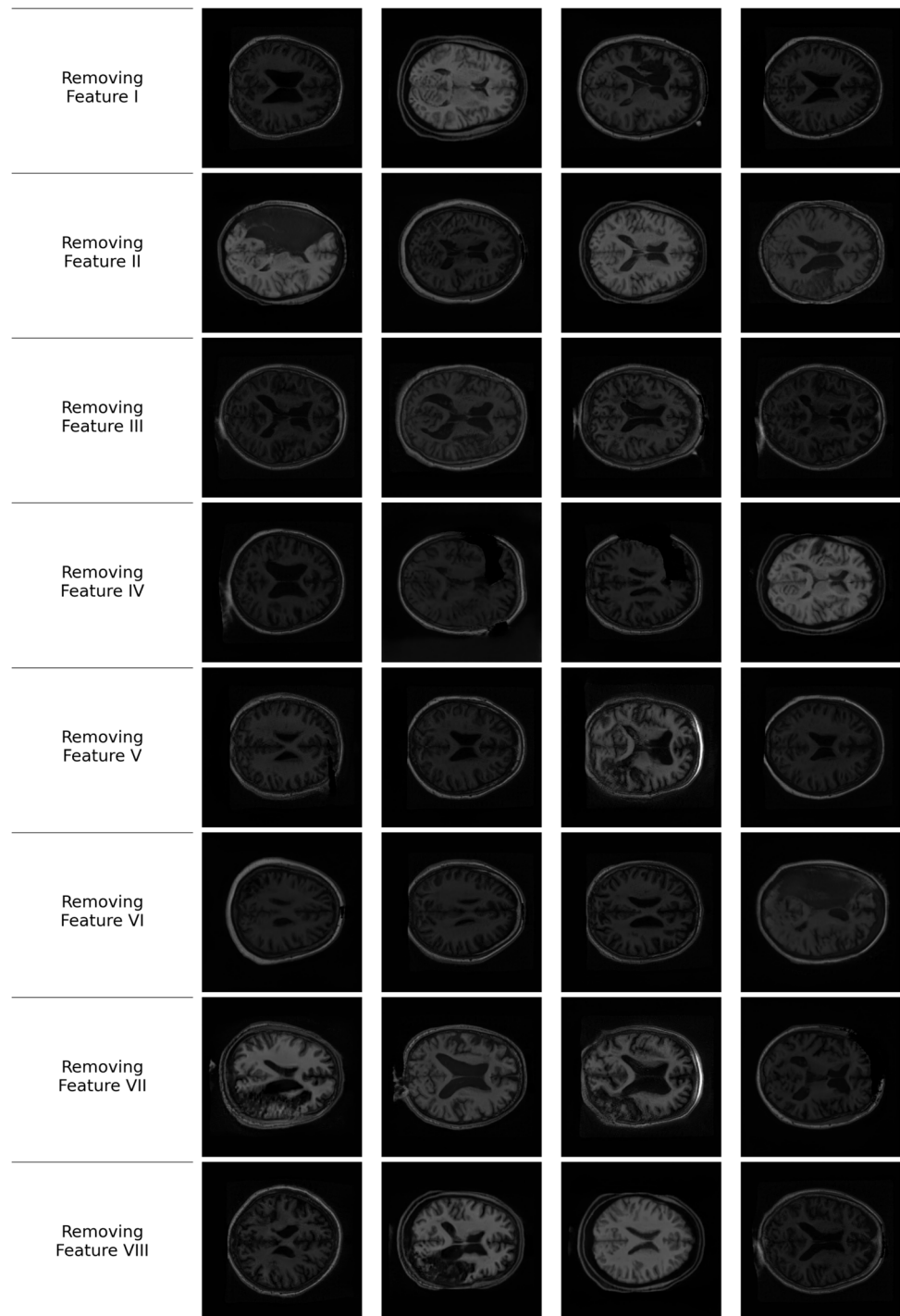
Targeting an improved result with a combined approach, $UncScore_{corr} < 0.035$ was chosen as a threshold so that any high-quality image classified by the Isolation Forest with $UncScore_{corr} \geq 0.035$ was also labeled as a low-quality image. The final combined result of the one-class classifier plus the $UncScore_{corr}$ filter was that 19.52% of the synthetic images were classified as anomalies. Figures 13 and 14 show a random sample of the synthetic images in high- and low-quality classes with the combined approaches. It is possible to observe that both dubious images and masks were labeled as low-quality, while all the images in the high-quality class have good visual quality.

Then, qualitative metrics were used in order to verify the difference between the set of real images with different arrangements of synthetic images. For simplicity, Table 4 defines the configurations of image sets to be

Table 3 Evaluation of the features by removing them from the one-class classification. Each row shows the count of images that were classified as anomalies when using all the features to train the one-class classifier but were not classified when the referred feature was removed

Feature removed	Count of images
Feature I: Brain area	102
Feature II: Circumscribed rectangle orientation	297
Feature III: Perimeter	70
Features IV: Centroid X	34
Features V: Centroid Y	55
Feature VI: Intersecting ratio of the convex hull	74
Feature VII: Intersecting ratio of the background convex hull	40
Feature VIII: Circularity	116

Fig. 11 Examples of images that would not be classified as anomalies if each of the features were individually removed



used in this analysis, while the result of the metrics for each image set is presented in Table 5. The goal was to compare the similarity between the real images with all the synthetic images and three other configurations of image filtration. The Fréchet inception distance (FID) [41, 42] and its variant, the Kernel inception distance (KID), which is claimed to be more robust against an imbalanced number of images in the comparing datasets [43] were employed for that. Table 5 also brings the Inception Score (IS) [44] as a measure of the diversity of each image set and the number of images contained in each set to provide more context.

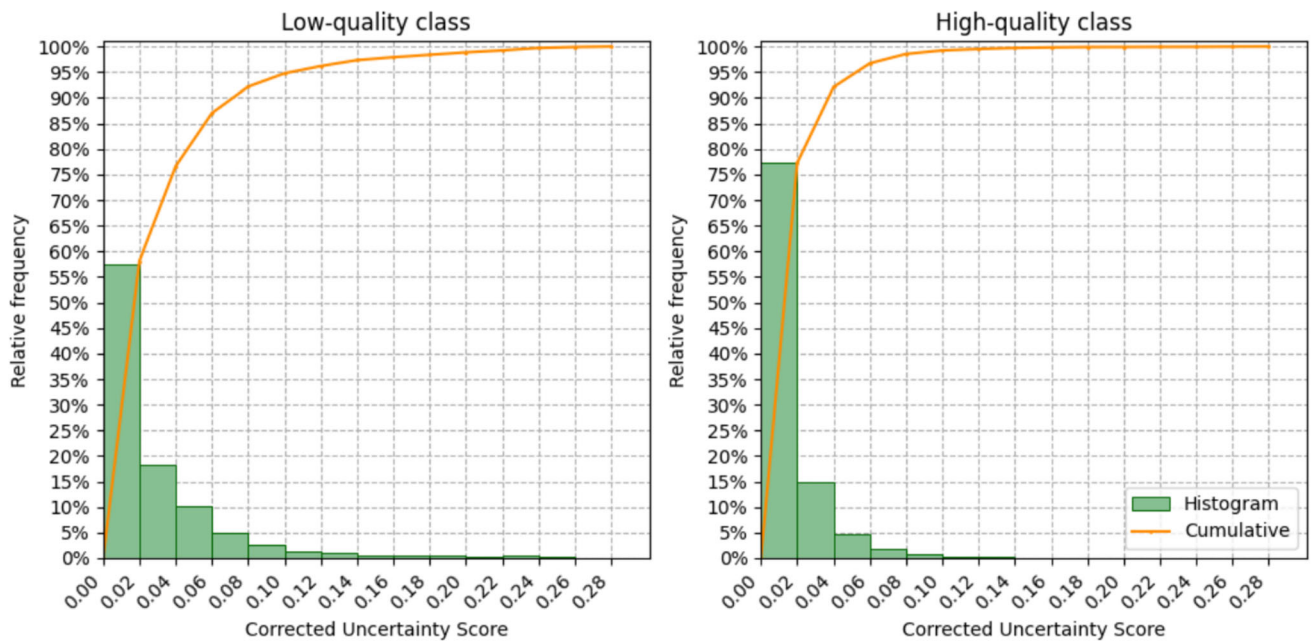


Fig. 12 Histograms of the corrected uncertainty score measurements for the low-quality and high-quality classes outputted by Isolation Forest model

Considering the FID, it is possible to observe that H_1 is the subset that is the most similar to the real images set (R), and both H_1 and H_3 are more similar to the real images than the image set containing all the synthetic images (S). On the other hand, considering KID, S , H_1 and H_3 have an equivalent similarity with R . These observations prove that our anomaly detection scheme was successful in correctly detecting images synthesized poorly. It is also expectable that H_1 would be slightly more similar to R than H_2 and H_3 since the removal of images with high uncertainty score is not perfectly bounded to the quality of the image synthesis, but with the quality of the segmentation mask—although we have proved some level of relationship through Figure 12 and the one-tailed Mann–Whitney U Test. It means that the final combined filtering strategy proposed (H_3) could improve the quality of the segmentation masks without significantly impacting the similarity with the real dataset. The IS results support this affirmation, where it is possible to observe that the diversity of the images is quite similar between R , S , H_1 , and H_3 .

Conversely, the sets containing anomalous images (L_1 , L_2 , and L_3) have much higher FID and KID values than H_1 , H_2 , and H_3 , confirming that the images removed by the filtering approaches are indeed outliers.

Another important consideration is that, in general, higher IS values relate to better results. However, in this study, this is not exactly the case because the sets containing anomalous images should have more image diversity and, thus, a higher IS. On the other hand, IS values much lower than those from the set of real images would indicate that the filtered sets would lose diversity if compared with the real set, which is not the case either.

4.3 Prevent the generation of low-quality images by *a priori* selection of latent vectors

We also considered whether the generation of lower-quality images could be avoided in the first place by *a priori* selecting only the latent vectors that, after feeding to StyleGAN2-ADA, would generate high-quality images. To answer this question, a multilayer perceptron (MLP) was trained to classify the latent codes that generated the synthetic images. The labels used (high-quality and low-quality) were the outputs of the Isolation Forest model without considering the uncertainty score measurements.

The MLP architecture has 512 input features fully connected to 128 neurons with a hyperbolic tangent ($\tanh$) activation, followed by another fully connected layer with 32 neurons with a $\tanh$ activation, which is connected

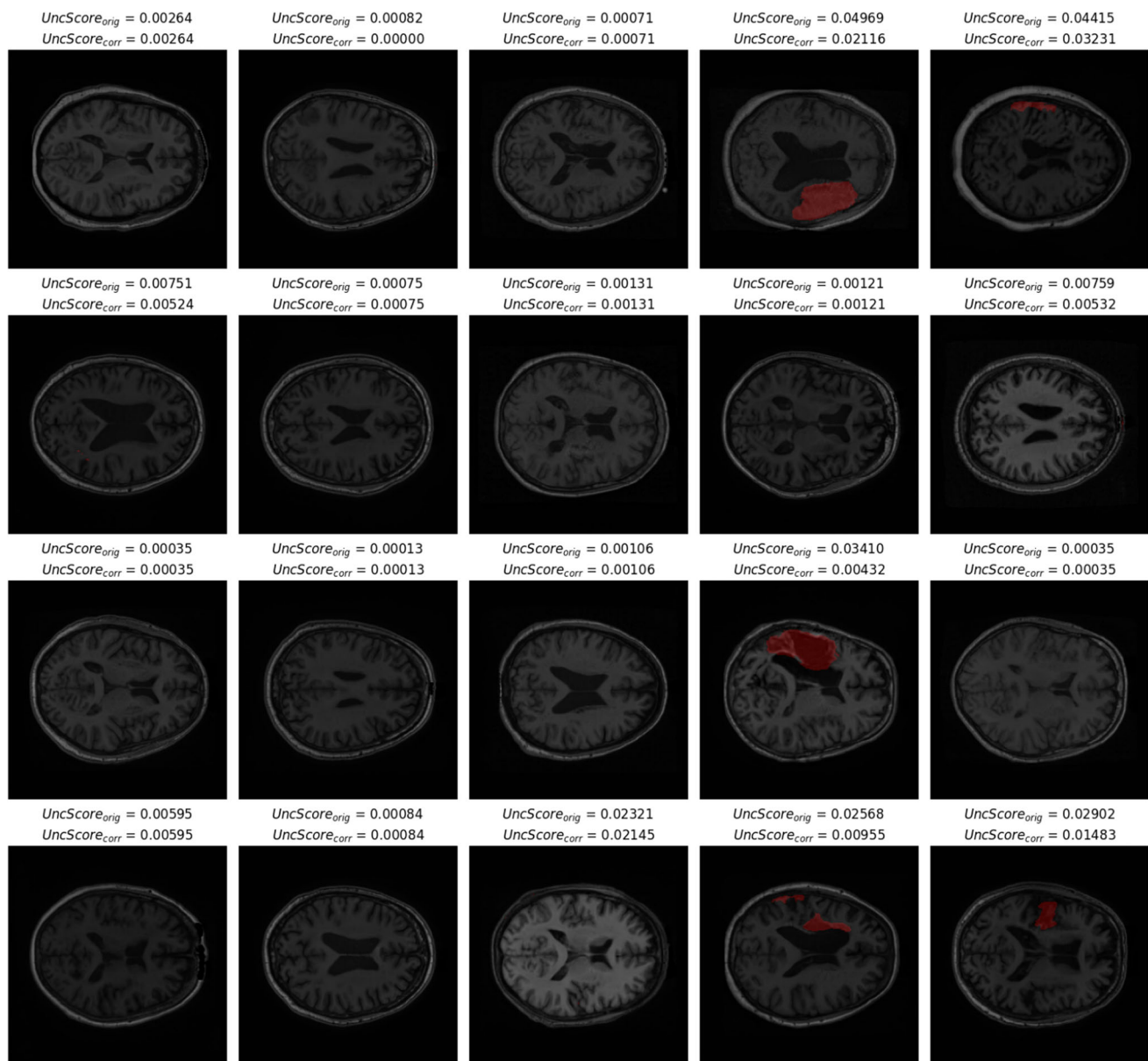


Fig. 13 Random sample of high-quality images filtered *a posteriori* of the synthetic images generated by DatasetGAN2-ADA

to an output layer containing a single neuron with a sigmoid activation. The output layer yields values between 0 and 1, which are rounded to make the final result binary (0—low-quality class, 1—high-quality class).

Two loss functions that are robust against class imbalance and are able to provide control over the precision–recall trade-off were evaluated. The first is the binary cross-entropy (BCE) with logits, represented by the Equation 3 for a single label.

$$l_i = -\omega_p y_i \log(p(y_i)) - (1 - y_i) \log(1 - p(y_i)) \quad (3)$$

Where i is the image's latent code index, y_i is the real label for the latent code i , $p(y_i)$ is the output of the sigmoid layer before rounding, and ω_p is the positive weight defined by equation 4. The higher the weight ω , the higher the expected recall and the lower the expected precision. However, as the values of ω get closer to zero, the precision tends to be higher at the cost of lower recall and possibly lower accuracy.

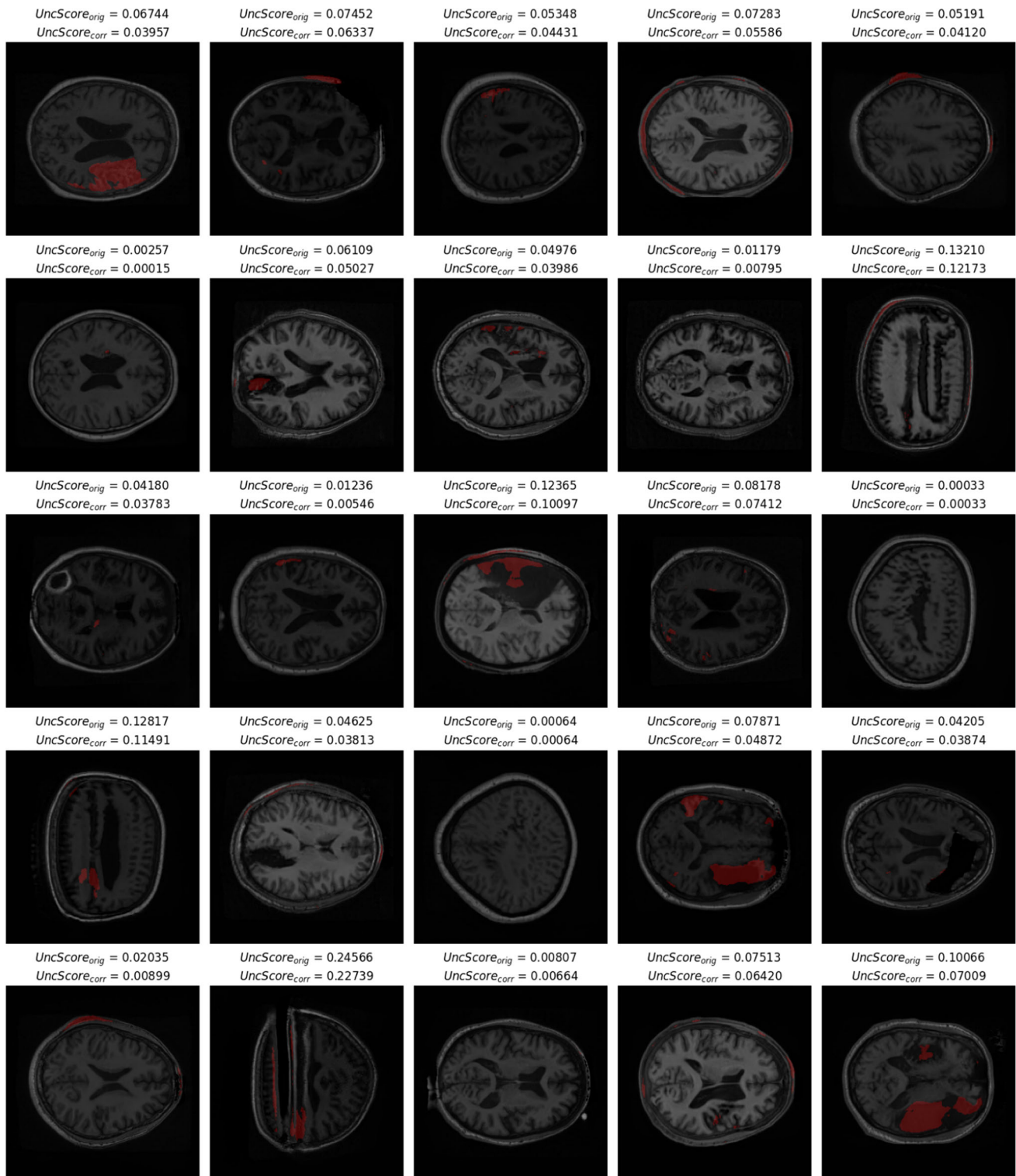


Fig. 14 Random sample of low-quality images filtered *a posteriori* of the synthetic images generated by DatasetGAN2-ADA

$$\omega_p = \omega.Prevallence \quad (4)$$

The second loss function evaluated is the Focal Loss [45], which might be understood in this case as a generalization of BCE with logits when $\gamma = 0$, as shown in Equation 5.

Table 4 Definition of image sets

Image set	Definition
R	All real images
S	All synthetic images
L_1	Anomalous images from the one-class classification
H_1	$S - L_1$
U_1	All the synthetic images with $UncScore_{orig} \geq 0.035$
U_2	All the synthetic images with $UncScore_{corr} \geq 0.035$
L_2	$L_1 \cup U_1$
H_2	$S - L_2$
L_3	$L_1 \cup U_2$
H_3	$S - L_3$

Table 5 FID and KID between the real images (R) and subsets of the synthetic images alongside the corresponding Inception Score and size of each image set

Image set	FID	KID	IS (mean, std)	Subset size
R	N/A	N/A	(1.495, 0.012)	13,100
S	4.969	0.001	(1.502, 0.007)	10,000
L_1	20.150	0.012	(1.708, 0.052)	1,061
H_1	4.722	0.001	(1.471, 0.007)	8,939
L_2	12.813	0.006	(1.643, 0.032)	2,762
H_2	5.207	0.002	(1.440, 0.011)	7,238
L_3	15.844	0.008	(1.680, 0.033)	1,952
H_3	4.855	0.001	(1.450, 0.007)	8,048

$$l_i = -\alpha(1 - p(y_i))^\gamma y_i \log(p(y_i)) - (1 - \alpha)p(y_i)^\gamma (1 - y_i) \log(1 - p(y_i)) \quad (5)$$

Where the variation of α has a similar effect in the Focal loss to the variation of ω_p in the BCE with logits loss, and γ aims to penalize the loss on misclassified examples.

The MLP was trained with the latent vectors as features and the resulting classes of the Isolation Forest model as targets. Different loss functions were evaluated, where in each trial, the training followed a tenfold cross-validation scheme for 300 epochs. The best epoch (E_{best}) of each of the ten iterations of the cross-validation scheme was chosen as the balance between validation accuracy and validation precision according to Equation 6 for the BCE with logits loss, while for the Focal loss selected as the epoch with minimal validation loss.

The hyperparameters were chosen so that all combinations of $\omega_1 = \{0, 0.5, 1, 2\}$ and $\omega = \{0.01, 0.1, 0.25, 0.5, 1, 2, 10, 100\}$ were evaluated for BCE with logits loss. On the other hand, for the focal loss, all combinations of $\alpha = \{1, 0.75, 0.5, 0.25, 0.01\}$ and $\gamma = \{0.5, 1, 2, 5\}$ were assessed.

$$E_{best} = \arg \max_{e \in \{1, 2, \dots, 300\}} \left(\frac{Accuracy_e + \omega_1 \cdot Precision_e}{2} \right) \quad (6)$$

Additionally, the results were assessed before and after the *a posteriori* filtration of $UncScore_{corr}$ in the cross-validation scheme. The former is simply the mean result over the ten cross-validation iterations. The latter was calculated by storing the classification outcomes of the validation images and their $UncScore_{corr}$ for each of the ten iterations in the cross-validation scheme. Then, after applying the $UncScore_{corr}$ filter, the classifier metrics were corrected according to the final high- and low-quality classes.

Figure 15 shows a precision–recall plot of the results on its top half and a plot comparing precision and accuracy on its bottom half. The best results are in bold in Tables 6 and 7.

The results are remarkable, especially because the MLP achieved high precision, which means high performance in eliminating low-quality images. However, it is important to notice that the prevalence of the high-quality class is 89.39%. In other words, we expect that by selecting the latent vectors prior to image synthesis, in the best-case scenario shown in Table 7, it would produce 95.24% of good-quality images instead of 89.39% before the $UncScore_{corr}$ filtration. It represents an improvement of almost six percentage points and indicates that using the latent vector classifier *a priori* of the image synthesis is a suitable alternative.

Furthermore, the employment of the latent vector classifier *a priori* of the image synthesis allows performing anomaly detection with a controlled number of elements, avoiding the need to scale it to a huge number of images, which could possibly make the one-class classification computationally more expensive or even prohibitive. Additionally, it allows an improved workflow as the model will not spend time and resources generating images with poor quality for *a posteriori* filtration in a separate process.

Fig. 15 Mean validation performance of the latent code classifier under different arrangements displayed in the format of a precision–recall plot on the top and a precision–accuracy plot on the bottom

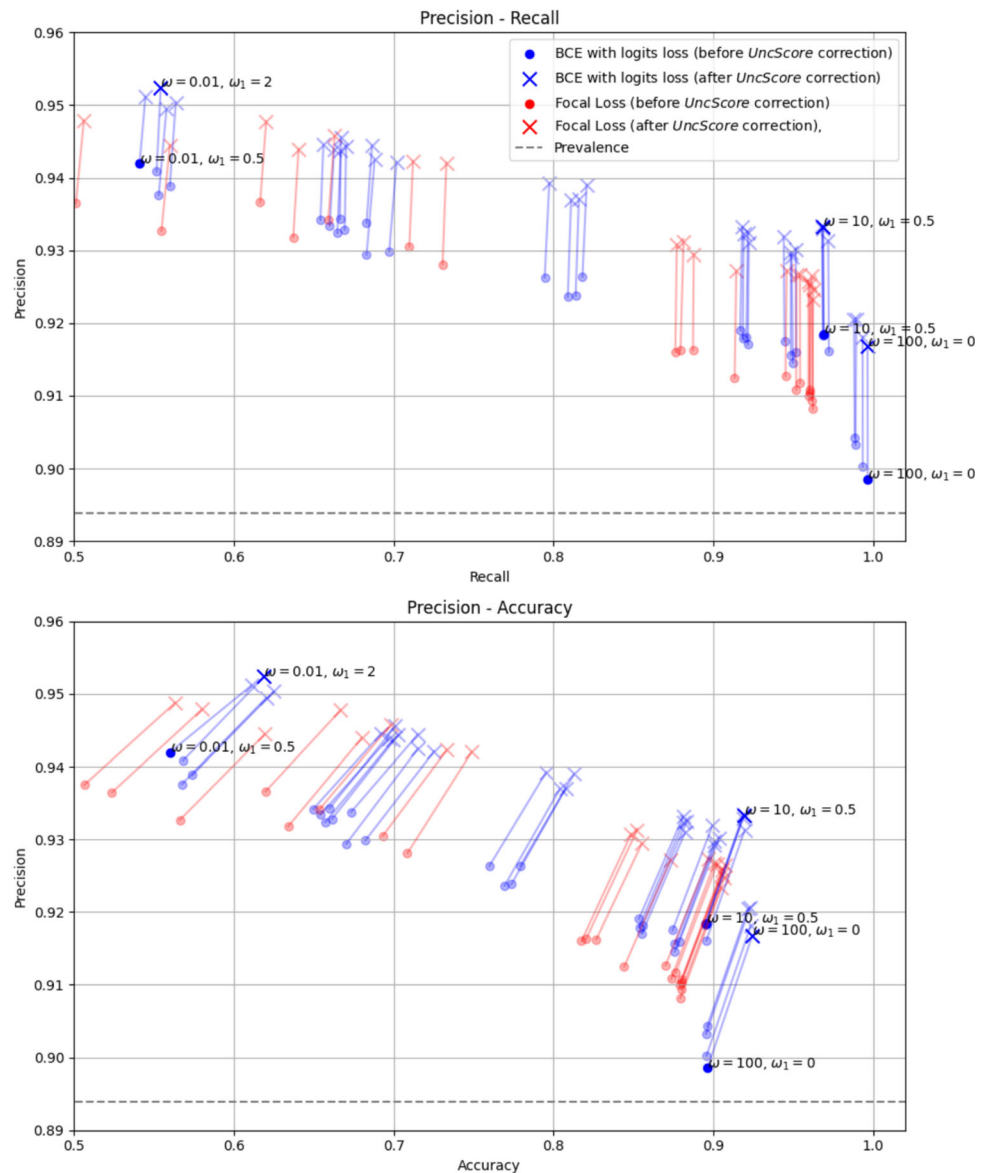


Table 6 Mean validation results of the latent vector classifier before the filtration of images with high $UncScore_{corr}$

	Loss function	Hyperparameters	Accuracy	Precision	Recall
Best precision	BCE with logits	$\omega = 0.01, \omega_1 = 0.5$	0.5600	0.9420	0.5411
Best accuracy	BCE with logits	$\omega = 100, \omega_1 = 0$	0.8963	0.8985	0.9965
Best balance	BCE with logits	$\omega = 10, \omega_1 = 0.5$	0.8954	0.9185	0.9690

Table 7 Mean validation results of the latent vector classifier after the filtration of images with high $UncScore_{corr}$

	Loss function	Hyperparameters	Accuracy	Precision	Recall
Best precision	BCE with logits	$\omega = 0.01, \omega_1 = 2$	0.6190	0.9524	0.5543
Best accuracy	BCE with logits	$\omega = 100, \omega_1 = 0$	0.9243	0.9168	0.9964
Best balance	BCE with logits	$\omega = 10, \omega_1 = 0.5$	0.9190	0.9333	0.9686

4.4 Potential challenges and limitations

Although the results achieved are remarkable for the ATLAS 2.0 dataset, it is important to consider that they were obtained for the specific case of MRI slices of human brains. Employing them to solve other problems might require adjustments, especially in the features conceived for detecting anomalies, which will differ depending on the problem domain. Likewise, the level of image diversity expected for successfully generating synthetic slices of human brains might differ for other tasks.

Additionally, the correction of the uncertainty score measures was quite successful for the dataset studied, but it is expected that the bias would change for different datasets, as the imbalance of the segmentation classes will most certainly vary. Thus, adjusting the bias is an important requirement for further work with different datasets.

Furthermore, the prevention of the synthesis of low-quality images by *a priori* classification of the latents was conceived for this specific problem. Using it in other datasets requires training the MLP again, considering that the success of the training is bound to the quality of the anomaly detection model.

5 Conclusion

In this paper, we introduced DatasetGAN2-ADA, an enhanced deep-style interpreter designed for the automatic generation of annotated medical images. This model proposes to substantially improve the quality of synthesized medical images by presenting modifications in DatasetGAN and employing additional filtering to remove low-quality images and segmentation masks. Firstly, we upgraded DatasetGAN's backbone from StyleGAN to StyleGAN2-ADA, guaranteeing a superior quality of the synthesized images even when using smaller datasets. Secondly, we replaced the original DatasetGAN style interpreter with a revamped architecture which proved to reach better performance. Thirdly, we eliminated the dependency of manual annotations by employing image projections matched with preexisting semantics to train DatasetGAN2-ADA. Then, we employed an enhanced latent vector generator which avoids the synthesis of low-quality images by assessing the latent vectors before image synthesis. All these modifications were employed using an improved computational strategy that enables the use of DatasetGAN2-ADA with much lower RAM requirements than DatasetGAN.

Beyond providing these advancements, our method incorporates a novel anomaly detection technique to ensure the generation of high-fidelity medical images. The presented anomaly detection framework works on two fronts. The first avoids generating low-quality images by embedding a one-class classifier trained on extracted features to eliminate anomalous images. The second prevents the synthesis of anomalous masks by enhancing the mask's

uncertainty score measure proposed by DatasetGAN and filtering out those masks above an uncertainty score threshold.

The experimental results using images extracted from ATLAS 2.0 dataset demonstrate that the anomaly detection framework effectively improves the fidelity of the images and semantics synthesized, discarding up to 19.52% of the generated images due to quality issues. Furthermore, the latent vector classifier used to prevent the generation of latent vectors corresponding to anomalous images before image synthesis achieved up to 95.24% precision. These achievements highlight DatasetGAN2-ADA's potential for practical applications in medical imaging and beyond.

Future work could focus on further optimizing the latent space correlation with synthesis quality, exploring additional features to enhance the robustness and accuracy of the proposed system, and studies with datasets from other domains.

Acknowledgements This work is partially supported by the Autonomous Government of Andalusia (Spain) under project UMA20-FEDERJA-108, project name Detection, characterization and prognosis value of the non-obstructive coronary disease with deep learning, and also by the Ministry of Science and Innovation of Spain, grant number PID2022-136764OA-I00, project name Automated Detection of Non Lesional Focal Epilepsy by Probabilistic Diffusion Deep Neural Models. It includes funds from the European Regional Development Fund (ERDF). It is also partially supported by the University of Málaga (Spain) under grants B1-2021 20, project name Detection of coronary stenosis using deep learning applied to coronary angiography; B4-2023 13, project name Intelligent Clinical Decision Support System for Non-Obstructive Coronary Artery Disease in Coronarographies; B1-2022 14, project name Detección de trayectorias anómalas de vehículos en cámaras de tráfico, and B1-2023 18, project name Sistema de videovigilancia basado en cámaras itinerantes robotizadas; and, by the Fundación Unicaja under project PUNI-003 2023, project name Intelligent System to Help the Clinical Diagnosis of Non-Obstructive Coronary Artery Disease in Coronary Angiography. The authors thankfully acknowledge the computer resources, technical expertise and assistance provided by the SCBI (Supercomputing and Bioinformatics) center of the University of Málaga. They also gratefully acknowledge the support of NVIDIA Corporation with the donation of a RTX A6000 GPU with 48Gb. The authors also thankfully acknowledge the grant of the Universidad de Málaga and the Instituto de Investigación Biomédica de Málaga y Plataforma en Nanomedicina-IBIMA Plataforma BIONAND. The authors thankfully acknowledge the funding for open access charge to the University of Málaga / CBUA.

Author Contributions All authors contributed to the study's conception and design. Material preparation and experimentation was performed Marcos Sergio Pacheco dos Santos Lima Junior and supervised by Juan Miguel de Ortiz-de-Lazcano-Lobato and Ezequiel López-Rubio. The analysis of the results was performed by all authors. The manuscript was written by Marcos Sergio Pacheco dos Santos Lima Junior and José David Fernández-Rodríguez, and all authors contributed to its corrections. All authors read and approved the final manuscript.

Funding Funding for open access publishing: Universidad de Málaga/CBUA.

Data availability ATLAS 2.0 dataset [27] is available in a public repository accessible with the DOI link provided. The processed dataset will be available on request. The source code is available online at <https://github.com/icai-uma/DatasetGAN2-ADA>.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All the MRIs used in this work had any potentially identifying non-brain data, such as facial voxels or user information removed by the provider of the ATLAS 2.0 dataset, guaranteeing the full anonymization of MR brain images. Also, informed consent was obtained from all subjects by the provider of the ATLAS 2.0 dataset.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Celard P, Iglesias EL, Sorribes-Fdez JM, Romero R, Vieira AS, Borrajo L (2023) A survey on deep learning applied to medical images: from simple artificial neural networks to generative models. *Neural Comput Appl* 35(3):2291–2323. <https://doi.org/10.1007/s00521-022-07953-4>
2. Abdou MA (2022) Literature review: efficient deep neural networks techniques for medical image analysis. *Neural Comput Appl* 34(8):5791–5812. <https://doi.org/10.1007/s00521-022-06960-9>
3. Mumuni A, Mumuni F, Gerrar NK (2024) A survey of synthetic data augmentation methods in machine vision. *Mach Intell Res* 21(5):831–869. <https://doi.org/10.1007/s11633-022-1411-7>
4. Garcea F, Serra A, Lamberti F, Morra L (2023) Data augmentation for medical imaging: a systematic literature review. *Comput Biol Med* 152:106391. <https://doi.org/10.1016/j.compbiomed.2022.106391>
5. Foucart A, Debeir O, Decaestecker C (2023) Shortcomings and areas for improvement in digital pathology image segmentation challenges. *Comput Med Imaging Graph* 103:102155. <https://doi.org/10.1016/j.compmedimag.2022.102155>
6. Aggarwal R, Sounderajah V, Martin G, Ting DSW, Karthikesalingam A, King D, Ashrafian H, Darzi A (2021) Diagnostic accuracy of deep learning in medical imaging: a systematic review and meta-analysis. *NPJ Digital Med* 4(1):65. <https://doi.org/10.1038/s41746-021-00438-z>
7. Yu X, Wang J, Hong Q-Q, Teku R, Wang S-H, Zhang Y-D (2022) Transfer learning for medical images analyses: a survey. *Neurocomputing* 489:230–254. <https://doi.org/10.1016/j.neucom.2021.08.159>
8. Ayana G, Dese K, Abagaro AM, Jeong KC, Yoon S-D, Choe S (2024) Multistage transfer learning for medical images. *Artif Intell Rev* 57(9):232. <https://doi.org/10.1007/s10462-024-10855-7>
9. Goceri E (2023) Medical image data augmentation: techniques, comparisons and interpretations. *Artif Intell Rev* 56(11):12561–12605. <https://doi.org/10.1007/s10462-023-10453-z>
10. Makhoulouf A, Maayah M, Abughanam N, Catal C (2023) The use of generative adversarial networks in medical image augmentation. *Neural Comput Appl* 35(34):24055–24068. <https://doi.org/10.1007/s00521-023-09100-z>
11. Wu X, Li Z, Tao C, Han X, Chen Y-W, Yao J, Zhang J, Sun Q, Li W, Liu Y, Guo Y (2024) DEA: data-efficient augmentation for interpretable medical image segmentation. *Biomed Signal Process Control* 89:105748. <https://doi.org/10.1016/j.bspc.2023.105748>
12. Chen Y, Yan Z, Zhu Y (2024) A comprehensive survey for generative data augmentation. *Neurocomputing* 600:128167. <https://doi.org/10.1016/j.neucom.2024.128167>
13. Chen Y, Yang X-H, Wei Z, Heidari AA, Zheng N, Li Z, Chen H, Hu H, Zhou Q, Guan Q (2022) Generative adversarial networks in medical image augmentation: a review. *Comput Biol Med* 144:105382
14. Sreelekshmi V, Jyothisha JN (2024) Variational auto encoders for improved breast cancer classification. *Proc Comput Sci* 233:801–811
15. Srivastava R, Kumar P (2024) Deep-GAN: an improved model for thyroid nodule identification and classification. *Neural Comput Appl* 36(14):7685–7704. <https://doi.org/10.1007/s00521-024-09492-6>
16. Kugelman J, Alonso-Caneiro D, Read SA, Vincent SJ, Collins MJ (2024) Enhancing OCT patch-based segmentation with improved GAN data augmentation and semi-supervised learning. *Neural Comput Appl*. <https://doi.org/10.1007/s00521-024-10044-1>
17. Jain V, Nankar O, Jerrish DJ, Gite S, Patil S, Kotecha K (2021) A novel AI-based system for detection and severity prediction of dementia using MRI. *IEEE Access* 9:154324–154346
18. Rais K, Amroune M, Haouam MY (2024) Medical image generation techniques for data augmentation: Disc-vae versus gan. In: 2024 6th International Conference on Pattern Analysis and Intelligent Systems (PAIS), pp. 1–8. IEEE
19. Xun S, Li D, Zhu H, Chen M, Wang J, Li J, Chen M, Wu B, Zhang H, Chai X et al (2022) Generative adversarial networks in medical image segmentation: a review. *Comput Biol Med* 140:105063
20. Yu X, Li G, Lou W, Liu S, Wan X, Chen Y, Li H (2023) Diffusion-based data augmentation for nuclei image segmentation. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 592–602. Springer
21. Akrouf M, Gyepesi B, Holló P, Poór A, Kincsó B, Solis S, Cirone K, Kawahara J, Slade D, Abid L, et al. (2023) Diffusion-based data augmentation for skin disease classification: Impact across original medical datasets to fully synthetic images. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 99–109. Springer
22. Park Y-H, Kwon M, Choi J, Jo J, Uh Y (2023) Understanding the latent space of diffusion models through the lens of riemannian geometry. *Adv Neural Inf Process Syst* 36:24129–24142
23. Fernandes JND, Cardoso VEM, Comesaña-Campos A, Pinheira A (2024) Comprehensive review: machine and deep learning in brain stroke diagnosis. *Sensors* 24(13):4355. <https://doi.org/10.3390/s24134355>
24. Luo J, Dai P, He Z, Huang Z, Liao S, Liu K (2024) Deep learning models for ischemic stroke lesion segmentation in medical images: a survey. *Comput Biol Med* 175:108509. <https://doi.org/10.1016/j.compbiomed.2024.108509>

25. Zhang Y, Ling H, Gao J, Yin K, Lafleche J-F, Barriuso A, Torralba A, Fidler S (2021) Datasetgan: Efficient labeled data factory with minimal human effort. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10145–10155
26. Karras T, Aittala M, Hellsten J, Laine S, Lehtinen J, Aila T (2020) Training generative adversarial networks with limited data. *Adv Neural Inf Process Syst* 33:12104–12114
27. Liew S-L, Lo BP, Donnelly MR, Zavaliangos-Petropulu A, Jeong JN, Barisano G, Hutton A, Simon JP, Juliano JM, Suri A et al (2022) A large, curated, open-source stroke neuroimaging dataset to improve lesion segmentation algorithms. *Sci Data* 9(1):320. <https://doi.org/10.1038/s41597-022-01401-7>
28. Karras T, Laine S, Aila T (2019) A style-based generator architecture for generative adversarial networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 4401–4410
29. Li D, Ling H, Kim SW, Kreis K, Fidler S, Torralba A (2022) Bigdatasetgan: Synthesizing imagenet with pixel-wise annotations. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 21330–21340
30. Ling H, Kreis K, Li D, Kim SW, Torralba A, Fidler S (2021) Editgan: high-precision semantic image editing. *Adv Neural Inf Process Syst* 34:16331–16345
31. Karras T, Laine S, Aittala M, Hellsten J, Lehtinen J, Aila T (2020) Analyzing and improving the image quality of stylegan. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8110–8119
32. Elfving S, Uchibe E, Doya K (2018) Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks* 107, 3–11 <https://doi.org/10.1016/j.neunet.2017.12.012>. Special issue on deep reinforcement learning
33. Abdal R, Qin Y, Wonka P (2019) Image2stylegan: How to embed images into the stylegan latent space? In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 4432–4441
34. Abdal R, Qin Y, Wonka P (2020) Image2stylegan++: How to edit the embedded images? In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8296–8305
35. Fan Z, Kelkar V, Anastasio MA, Li H (2022) Application of datasetgan in medical imaging: preliminary studies. In: *Medical Imaging 2022: Image Processing*, vol. 12032, pp. 452–458. SPIE
36. Haralick R (1974) A measure for circularity of digital figures. *Systems, Man and Cybernetics, IEEE Transactions on SMC-4*, 394–396 <https://doi.org/10.1109/TSMC.1974.5408463>
37. Liu FT, Ting KM, Zhou Z-H (2008) Isolation forest. In: 2008 Eighth IEEE International Conference on Data Mining, pp. 413–422. <https://doi.org/10.1109/ICDM.2008.17>
38. Samariya D, Thakkar A (2023) A comprehensive survey of anomaly detection algorithms. *Ann Data Sci* 10(3):829–850
39. Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. In: Proceedings of the 31st International Conference on Neural Information Processing Systems. NIPS'17, pp. 4768–4777. Curran Associates Inc., Red Hook, NY, USA
40. King AP, Eckersley RJ (2019) Chapter 6 - inferential statistics III: Nonparametric hypothesis testing. In: King AP, Eckersley RJ (eds) *Statistics for Biomedical Engineers and Scientists*. Academic Press, London
41. Heusel M, Ramsauer H, Unterthiner T, Nessler B, Hochreiter S (2017) Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems* 30
42. Seitzer M (2020) pytorch-fid: FID Score for PyTorch. <https://github.com/mseitzer/pytorch-fid>. Version 0.3.0
43. Bińkowski M, Sutherland DJ, Arbel M, Gretton A (2018) Demystifying mmd gans. *arXiv preprint arXiv:1801.01401*
44. Salimans T, Goodfellow I, Zaremba W, Cheung V, Radford A, Chen X (2016) Improved techniques for training gans. *Advances in neural information processing systems* 29
45. Lin T-Y, Goyal P, Girshick R, He K, Dollár P (2017) Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2980–2988

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Marcos Sergio Pacheco dos Santos Lima Junior¹ 

Juan Miguel Ortiz-de-Lazcano-Lobato^{1,2,3}  · José David Fernández-Rodríguez^{1,2,3} 

Ezequiel López-Rubio^{1,2,3} 

✉ Marcos Sergio Pacheco dos Santos Lima Junior
marcos.pacheco@uma.es

Juan Miguel Ortiz-de-Lazcano-Lobato
jmortiz@lcc.uma.es

José David Fernández-Rodríguez
josedavid@uma.es

Ezequiel López-Rubio
ezeqlr@lcc.uma.es

- ¹ Department of Computer Languages and Computer Science, University of Málaga, Bulevar Louis Pasteur, 35, Málaga, 29071 Andalucía, Spain
- ² Biomedical Research Institute of Málaga (IBIMA), C/Doctor Miguel Díaz Recio, 28, City, 29010 Andalucía, Spain
- ³ ITIS Software, University of Málaga, C/. Arquitecto Francisco, Peñalosa 18, Málaga, 29010 Andalucía, Spain